Classifying Adolescent Excessive Alcohol Drinkers

A Survey-Data Classification Adaptation

Course: UE24CS352A — Machine Learning **Institution:** PES University

1. Problem Statement

The original project topic concerns classifying adolescent excessive alcohol drinkers from functional magnetic resonance imaging (fMRI) data. Because the original preprocessed fMRI dataset was unavailable for this implementation, an alternative dataset was used with faculty approval. This implementation investigates whether student survey and demographic variables can distinguish records in a project-defined high alcohol-consumption category from a lower-consumption category.

2. Dataset Details

The implementation uses the Portuguese-language file from the **UCI Student Performance** dataset. It contains 649 records and 33 original columns covering student demographics, social circumstances, school-related information, and self-reported alcohol-consumption ratings. Dataset source: UCI Machine Learning Repository — Student Performance.

Dalc represents reported workday alcohol consumption and **Walc** represents reported weekend alcohol consumption; each is an ordinal rating from 1 to 5. These are survey ratings, not direct quantities of alcohol consumed or clinical assessments.

3. Target Definition and Preprocessing

A binary target, `high_consumption`, was constructed for this project: **Class 1** if Dalc ≥ 4 or Walc ≥ 4; otherwise **Class 0**. This is a project-defined threshold, not a validated clinical definition of excessive drinking. The resulting dataset contains 509 lower-consumption records (78.4%) and 140 high-consumption records (21.6%).

Dalc and Walc were excluded from the model inputs because they define the target, preventing direct target leakage. Numerical features were median-imputed and standardized; categorical features were imputed using the most frequent value and one-hot encoded. These operations were placed inside a scikit-learn pipeline so preprocessing is fitted within training folds.

Target class	Records	Percentage
Lower consumption (0)	509	78.4%
High consumption (1)	140	21.6%
Total	649	100%

Scope note: This is a survey-data adaptation, not an fMRI classifier. The model does not diagnose alcohol use disorder or establish individual clinical risk.

4. Approach and Implementation

Two baseline classifiers were evaluated: **Logistic Regression**, a linear classifier, and an **RBF-kernel Support Vector Machine (SVM)**, which can model nonlinear decision boundaries. Both used balanced class weights to account for class imbalance. The data were divided using a stratified 80/20 train/test split with random state 42. Five-fold stratified cross-validation on the training set used F1-score as its scoring metric. The held-out test set contained 130 records (102 lower-consumption and 28 high-consumption).

5. Results

Metric	Logistic Regression	RBF SVM
Accuracy	0.746	0.792
Balanced accuracy	0.735	0.738
Precision (high class)	0.444	0.514
Recall (high class)	0.714	0.643
F1-score (high class)	0.548	0.571
ROC-AUC	0.826	0.854
5-fold CV F1 mean ± SD	0.550 ± 0.030	0.527 ± 0.041

Confusion matrix (actual rows; predicted columns)	Predicted lower	Predicted high
LR — Actual lower	77	25
LR — Actual high	8	20
SVM — Actual lower	85	17
SVM — Actual high	10	18

6. Interpretation and Conclusion

On this held-out test split, RBF SVM achieved higher accuracy (79.2%), precision (0.514), F1-score (0.571), and ROC-AUC (0.854). Logistic Regression had higher recall for the high-consumption class (0.714 versus 0.643) and a slightly higher mean cross-validation F1-score. Logistic Regression identified 20 of 28 high-consumption test records; SVM identified 18. The results do not establish that one model is consistently superior.

The experiment demonstrates a working baseline survey-data classification pipeline. Its limitations include self-reported ratings, class imbalance, one held-out split, limited external validation, and uncertain generalization to other populations. Predictive associations do not establish causality. Future work could test alternative thresholds and additional validation; reproducing the original research would require appropriate fMRI data and participant-level labels.

7. Reproducibility

The project includes scripts for target preparation (`src/prepare_dataset.py`), training/evaluation (`src/train_baseline.py`), and visualization (`src/visualize_results.py`). With the UCI file saved as `data/student-por.csv`, run the following from the repository root:

```
source .venv/bin/activate
python src/prepare_dataset.py
python src/train_baseline.py
python src/visualize_results.py
```

References

1. UCI Machine Learning Repository. *Student Performance*. https://archive.ics.uci.edu/dataset/320/student%2Bperformance.

2. Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. Proceedings of the 5th Future Business Technology Conference (FUBUTEC).

3. Noh, J. et al. Original CS229 project materials, *Classifying Adolescent Excessive Alcohol Drinkers from fMRI Data*. https://github.com/jnoh4/CS229ToShare. Cited for original-topic context; the results above come from the alternative UCI survey dataset.